ADAM HEMMOUDA

COMPUTER ENGINEER | EMBEDDED SYSTEMS | TECHNICAL LEADERSHIP

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PROFILE

Computer engineering graduate and U.S. Army Signal Corps officer with hands-on experience in FPGA design, embedded systems, networked applications, electronics manufacturing, technical documentation, and cross-team leadership. Brings disciplined systems thinking to defense and aerospace engineering challenges.

EDUCATION

Rochester Institute of Technology May 2026
B.S. Computer Engineering Technology, *cum laude*

CORE CAPABILITIES

Embedded & Digital	C/C++, VHDL, FPGA development, microcontrollers, RTOS, state machines, HDL simulation
Signals & Electronics	DSP, sampling, digital filtering, RF PCB design, signal conditioning, A/D and D/A conversion
Software & Networking	Python, Java, Assembly, MATLAB, TCP/IP, SQL, LAN/WAN, system integration
Engineering Tools	Quartus II, Multisim, STM32CubeIDE, Code Composer Studio, Eclipse, Git/GitHub

SELECTED ENGINEERING WORK

FPGA Stereo Vision Tennis Ball Tracker 2025-2026
Senior Capstone

- Implemented a VHDL image-processing pipeline on a Snickerdoodle FPGA to threshold stereo images, detect centroids, calculate disparity, and reconstruct 3D position.
- Built a physics-based Blender test environment and used MATLAB to model stereo geometry; integrated Simulink over Wi-Fi and Python for camera, pipeline, and control logic.
- Presented formal preliminary and critical design reviews supported by system diagrams and engineering documentation.

EXPERIENCE

Second Lieutenant, Signal Corps Officer May 2026-Present
United States Army - Infantry Branch Detail

- Commissioned through RIT Army ROTC; as cadet Task Force OIC, led a 170+ person training operation across four ROTC programs.

Junior Manufacturing Engineer Oct 2025-Jun 2026
Simcona Electronics Corp.

- Developed a TCP/IP approval application backed by SQL, replacing manual paperwork with assigned signers, authenticated access, and email notifications.
- Prepared 13 wire lists and 26 equipment-label files with a four-person engineering team while resolving manufacturing and documentation discrepancies.
- Authored a 127-page illustrated technical manual covering configuration, assembly, and troubleshooting.

Control Panel Manufacturing & Inspection Co-op Sep 2024-May 2025
Simcona Electronics Corp.

- Built a CAD-modeled workstation prototype using 80/20 extrusion, mechanical hardware, and linear actuators.
- Created a Python label-printing tool using QR codes, panel-specific layouts, and XML commands to improve traceability.
- Manufactured and inspected four control-panel variants; established a 10-unit batch workflow and produced 50 units against a 20-unit weekly quota on one configuration.